



Empowering Nigerian MSMEs through Finance, Technology and Capability: The SCALE Framework for Sustainable Industrial Development

CHIA Shiaondo Kenneth

CHIATECH Solutions and Resources Ltd, Abuja, Nigeria

ORCID: 0009-0009-6434-4586 · Correspondence: chiakenneth01@gmail.com

Issue	CHIATECH JOURNAL · Volume 1 · Issue 1 · July 2026 collection · e018
Publication identifiers	ISSN not assigned—application ongoing · Full DOI not assigned—registration ongoing
Reserved DOI suffix	chj.g9d5lu · Opaque suffix reserved locally; registrar-supplied prefix will be added only after registration
Editorial status	Final editor's production copy · Public release requires recorded editorial and peer-review authorization
Evidence design	Policy and management review. No firms were surveyed and no administrative or financial records were analysed for this paper. Historical estimates retain their source dates.
Planned open licence	CC BY 4.0 upon authorized publication; no licence is asserted by this production copy alone

Abstract

Micro, small and medium enterprises are central to Nigerian livelihoods and production, yet policy discussion often treats finance, technology, management capability, markets and infrastructure as separate problems. This review synthesises official Nigerian enterprise statistics, finance-market studies, credit-infrastructure evidence, data-protection law and technology-adoption literature. It proposes the SCALE framework: Structured records and formalisation pathways; Capital suited to business need and cash flow; Adoption of useful and trustworthy technology; Linkages to customers, suppliers and support institutions; and Enabling infrastructure, capability and resilience. SCALE treats credit as one component of an enterprise system rather than a universal first intervention. It sequences basic transaction records and unit economics before more complex digital systems, aligns working-capital instruments with cash-conversion cycles, and links adoption decisions to measurable operating problems. It also addresses cyber risk, personal-data governance, gender and disability inclusion, business continuity and climate exposure. A staged implementation scorecard separates reach from use, business performance and causal impact. The paper does not update the 2017 national MSME census, estimate a current enterprise population or claim that the framework improves survival or employment. Published finance-gap estimates are reported with their study dates and definitions and should not be treated as interchangeable. SCALE is a testable management and policy architecture whose modules should be adapted by sector, firm maturity and location, then evaluated against explicit counterfactual and cost assumptions.

Keywords: MSMEs; Nigeria; access to finance; digital adoption; management capability; industrial development; business resilience

1. Introduction

MSME development programmes frequently begin with a preferred instrument—a loan, application, training course, grant or marketplace—and then search for businesses to receive it. Enterprises experience the problem differently. A retailer may need inventory finance but lack reliable sales records; a fabricator may have orders but face power and machine downtime; a food processor may need quality control and packaging before wider market access; a professional service firm may need contract discipline and cyber security rather than physical capital. A useful framework must diagnose the binding constraint and the firm's capacity to absorb an intervention.

The National Bureau of Statistics and Small and Medium Enterprises Development Agency of Nigeria reported 41,543,028 MSMEs in the 2017 national survey, of which 99.8% were classified as micro enterprises (NBS & SMEDAN, 2019). This is a dated survey estimate, not a 2026 count. The scale and



composition nevertheless underline why a single formalisation or finance pathway cannot fit all enterprises.

IFC studies have documented substantial unmet finance demand, low digital-financial-service use in surveyed segments and potential for supply-chain finance, while a later review found progress in Nigerian credit infrastructure through credit reporting, movable-asset security and the National Collateral Registry (IFC, 2022a, 2022b, 2024). These findings support a systems approach but do not establish that every firm should borrow or that credit alone causes growth.

2. Review Method and Evidence Discipline

The paper reviews the 2017 national MSME survey, IFC market and credit-infrastructure studies, the Central Bank of Nigeria's FinTech Report, the Nigeria Data Protection Act 2023 and management literature on digital transformation, dynamic capabilities and business-model development. Evidence is mapped to observable enterprise decisions and programme controls. The review is selective rather than systematic.

Published finance-gap numbers differ because they may cover formal or informal enterprises, a surveyed addressable market, total unconstrained demand or a product such as supply-chain finance. Currency values also embed date-specific prices and exchange rates. This paper therefore reports the source's measure and year and does not combine the approximately ₦13 trillion addressable opportunity in IFC's MSME Market Bite with the broader financing-gap estimate cited in its supply-chain study.

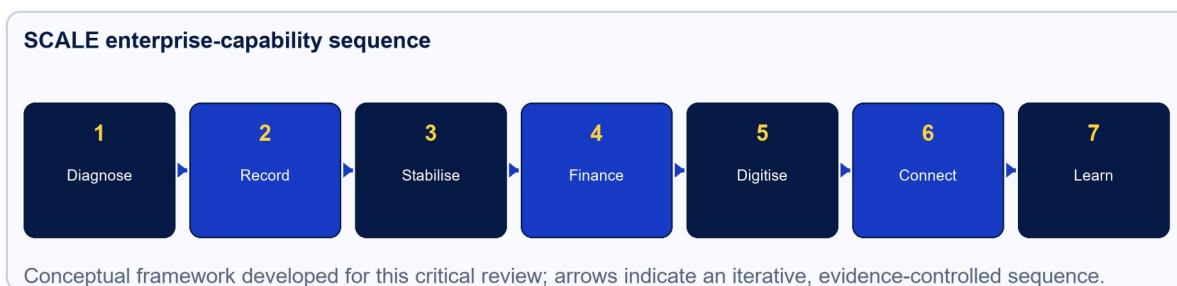


Figure 1. SCALE enterprise-capability sequence.

Note. Original conceptual diagram developed for this paper; the sequence is iterative and has not been validated as a measurement scale.

3. The SCALE Framework

Table 1. SCALE dimensions and enterprise evidence

Dimension	Management objective	Minimum evidence before support
Structured records	Make transactions, obligations and unit economics visible	Separated cash records, sales and cost evidence, inventory or service log
Capital suited to need	Match instrument, amount, tenor and repayment to cash flow	Use of funds, cash-conversion cycle, downside case and repayment source
Adoption of useful technology	Solve a measured operating problem safely and affordably	Baseline, user workflow, total cost, data and cyber controls
Linkages	Improve reliable access to customers, suppliers, anchors and support	Quality requirements, contracts, fulfilment capacity and complaint process
Enablers, capability and resilience	Sustain operations through skills, infrastructure and risk management	Capability gaps, continuity plan, insurance or risk treatment and owner

Note. Authors' synthesis unless otherwise stated; application requires context-specific evidence and professional judgment.



3.1 Structured records and proportionate formalisation

Records are an operating instrument before they are a compliance instrument. At minimum, the owner should separate household and business cash, record daily sales and expenses, track receivables and payables, reconcile cash and digital payments, and count critical inventory. These practices permit contribution-margin, break-even and cash-gap calculations. They also reduce dependence on memory and make finance applications more credible.

Formalisation should be sequenced to benefit and obligation. Registration, tax, sector licences, product standards, employment duties and data protection have different triggers. A programme should explain costs, rights and recurring obligations rather than promise that registration automatically unlocks finance. Assisted pathways are needed for low-literacy users, persons with disabilities and entrepreneurs operating in local languages.

3.2 Capital suited to business need and cash flow

Finance diagnosis begins with purpose. Inventory purchase, equipment, receivables, seasonal production and product development have different risk and repayment profiles. Short-tenor debt should not fund a long payback asset merely because it is available. Owners and lenders should model the cash-conversion cycle, repayment coverage under a downside case, currency and interest-rate exposure, collateral consequences and the cost of delayed customer payment.

Nigeria's credit infrastructure offers routes beyond land collateral, including credit reporting and security interests in movable assets. Supply-chain finance can use commercial relationships and receivables, although availability, documentation and pricing vary (IFC, 2022b, 2024). Equity, grants, savings, leasing, guarantees and blended instruments may be more appropriate at different stages. Responsible finance includes transparent pricing, no coercive collection and referral where borrowing would deepen distress.

3.3 Adoption of useful and trustworthy technology

Digitalisation should address an observed bottleneck. A payment tool may reduce cash leakage; simple inventory software may prevent stock-outs; customer-relationship records may improve follow-up; computer-aided design may reduce rework; and e-commerce may widen reach. Each use case needs a baseline, adoption owner, workflow redesign and a success threshold. Buying software without changing the work system produces subscription cost rather than capability.

The CBN FinTech Report emphasises innovation alongside trust, resilience, inclusion and risk management (CBN, 2025). MSMEs need basic cyber hygiene: supported devices, updates, multi-factor authentication, role-based access, backups, payment verification and an incident contact. Personal data should be collected for explicit purposes, minimised, protected, retained only as required and processed consistently with the Nigeria Data Protection Act 2023.

3.4 Linkages to markets and institutions

Market access is not achieved by listing a firm in a directory. A buyer relationship requires specifications, price, consistent quality, fulfilment, traceability, payment terms and a remedy for defects. Programme managers should help firms calculate the working-capital implications of a large order before celebrating the contract. Anchor firms can support supplier upgrading, but power asymmetry and late payment require transparent terms and complaint routes.

Cluster organisations, cooperatives, business-development providers, standards bodies, research institutions, financial institutions and logistics providers can reduce coordination costs when their roles



are clear. Referral should be closed-loop: the originating programme records whether the firm reached, used and benefited from the service, subject to lawful data-sharing and consent where applicable.

3.5 Enabling infrastructure, capability and resilience

Management capability converts resources into performance. Owners need practical competence in pricing, cost control, people management, quality, negotiation, procurement, digital risk and compliance. Training should be based on a diagnostic and produce an operating artefact—such as a cash forecast, standard operating procedure, customer register or continuity plan—followed by coaching and evidence of use.

Infrastructure risk should enter the business model. Electricity, water, connectivity, transport, security and waste services affect unit cost and delivery. Firms can map critical processes, acceptable downtime, dependencies, manual workarounds, backup power, alternate suppliers and recovery priorities. Climate and environmental risks should be considered alongside resource efficiency, worker safety and responsible waste handling.

4. Staged Adoption Path

Table 2. Staged enterprise support pathway

Stage	Entry condition	Priority actions	Exit evidence
1. See	Business activity exists but performance is opaque	Separate cash; record transactions; identify customer and value proposition	Four to twelve weeks of reconciled basic records
2. Stabilise	Records expose recurring operating failures	Fix pricing, leakage, quality, safety, stock and simple workflow controls	Stable process measure and documented routine
3. Finance	Specific viable use and repayment source are demonstrated	Compare instruments; model downside; prepare lender or investor evidence	Appropriate offer accepted or a documented no-borrow decision
4. Digitise	A measured process is ready for a tool	Select, configure, train, protect data and test business continuity	Active use and quantified process result
5. Connect	Quality and fulfilment capacity meet target market needs	Buyer qualification, contracting, logistics and supplier collaboration	Repeat order or verified market-learning result
6. Scale	Unit economics, controls and demand are repeatable	Delegate roles, standardise, invest and monitor risk	Sustained margin, cash and service performance

Note. Authors' synthesis unless otherwise stated; application requires context-specific evidence and professional judgment.

Stages are diagnostic, not a mandatory ladder. A mature firm may enter at finance or market linkage; a distressed firm may return to stabilisation. Programme incentives should reward correct referral and a justified no-investment decision. Enterprises should not be pushed into debt, software or formal procurement before minimum evidence exists.

5. Programme Design and Evaluation

A SCALE programme begins with segmentation by sector, maturity, owner goals, location and constraint—not by an undifferentiated label of MSME. The delivery team combines generalist case management with specialist finance, operations, market, technology and compliance support. Assistance is time-bounded and documented in an enterprise improvement plan with baseline, owner, milestone, cost and decision date.



Table 3. Measurement framework for SCALE programmes

Layer	Example measure	Claim permitted
Reach	Eligible firms enrolled by size, sector, gender, disability and location	Who entered; not who benefited
Use	Firms maintaining records or using the supported tool after 3 and 12 months	Adoption and persistence; not impact
Capability	Verified improvement in forecasting, quality control or continuity practice	Change in defined management practice
Performance	Sales, gross margin, cash cycle, defects, delivery and downtime	Observed change with uncertainty and source limits
Finance	Approvals, amount, effective cost, repayment and distress	Access and portfolio result; avoid survival bias
Impact	Incremental profit, survival or employment versus a credible comparison	Causal effect only with defensible design

Note. Authors' synthesis unless otherwise stated; application requires context-specific evidence and professional judgment.

Administrative dashboards should preserve denominators and distinguish firms contacted, eligible, enrolled, completing, adopting and remaining observable. Financial values need price-date treatment. Employment should use a consistent definition and avoid double-counting temporary workers. Independent evaluation should pre-specify outcomes and analyse attrition. Randomised phase-in, matched comparison or regression discontinuity may be possible depending on allocation rules; qualitative process research explains mechanisms and unintended effects.

Cost-effectiveness should include staff time, subsidy, technology, defaults, data systems and the entrepreneur's time. A programme that reaches fewer firms with durable practice change may outperform a high-volume seminar. Results should be reported by relevant subgroups without exposing individual firms or drawing conclusions from very small cells.

6. Risks, Limitations and Research Priorities

Record requirements can exclude informal or low-literacy enterprises if support is punitive. Digital footprints may expand surveillance or predatory lending. Marketplaces can create dependency, while anchor-led finance can transfer buyer risk to suppliers. Formalisation may increase costs before benefits. Loans can worsen distress; grants can distort selection; and short programmes may report only successful survivors. Programme governance needs complaints, conflicts-of-interest disclosure, responsible finance criteria and independent data checks.

This paper did not conduct a systematic review or collect firm-level data. The 2017 national survey is not a current enterprise census, and cited finance studies use different samples and definitions. SCALE has not been validated as a maturity score. Research should test inter-rater reliability, uptake burden and predictive validity, then compare modular support with standard business-development services. Sector-specific trials should measure net profit, cash flow, survival, job quality, productivity and owner welfare over sufficient time.

7. Conclusion

SCALE links the five conditions that allow enterprise support to become operational capability: visible records, suitable capital, useful and trustworthy technology, functioning market linkages, and enabling skills and resilience. Its central discipline is sequencing. Finance follows a viable use and repayment logic; digitalisation follows a defined process; market access follows quality and fulfilment readiness; scaling follows repeatable economics and controls. The framework offers a practical basis for piloting and evaluation, not a claim of proven impact or an updated count of Nigerian MSMEs.

**Declarations****Ethics and participant involvement**

Not applicable. This review/framework used publicly available documentary and scholarly sources. It involved no human participants, animals, patient material, identifiable records or intervention.

Data availability

No primary dataset was generated or analysed. The documentary and scholarly sources used are identified in the reference list; access conditions remain those of the source publishers.

Funding

Prepared within the publisher's journal-launch editorial research programme. No claim of grant-funded primary research is made.

Competing interests

CHIA TECH SOLUTIONS AND RESOURCES LIMITED is the corporate author and the publisher of CHIA TECH JOURNAL. This relationship is disclosed. Publication requires independent editorial handling and an auditable peer-review and authorization record.

Corporate author contribution and accountability

Corporate authorship is attributed to CHIA TECH SOLUTIONS AND RESOURCES LIMITED. An authorized representative must approve the final intellectual content and accept accountability before public release; this production file does not substitute for that recorded approval.

Generative AI disclosure

Generative AI assisted search planning, source triage, drafting support, language editing and Word-production automation. It was not treated as an author or source of evidence. Every citation, claim, legal reference and editorial decision requires accountable human verification before publication.

Peer-review and editorial status

This is a final editor's production copy for the July 2026 collection. It does not itself prove that external peer review, plagiarism screening, author approval, acceptance or publication authorization has occurred. Those records must be completed and retained in the journal's editorial system before release.

Figure and table provenance

The conceptual figure and synthesis tables were created specifically for this review. They contain no patient, participant, survey or third-party image data.

References

- Central Bank of Nigeria. (2025). CBN FinTech report: Policy Insight Series. https://www.cbn.gov.ng/Out/2026/CCD/CBN_FINTECH_REPORT_.pdf
- Federal Republic of Nigeria. (2023). Nigeria Data Protection Act, 2023. Nigeria Data Protection Commission. <https://ndpc.gov.ng/download/nigeria-data-protection-act-2023>
- International Finance Corporation. (2022a). Market Bite Nigeria: Innovation offers key to the broader MSME finance market. <https://www.ifc.org/en/insights-reports/2022/market-bite-nigeria-innovation-offers-key-to-the-broader-msme-finance-market>
- International Finance Corporation. (2022b). Supply chain finance market assessment: Nigeria. <https://www.ifc.org/en/insights-reports/2022/supply-chain-finance-market-assessment-nigeria>



- International Finance Corporation. (2024). Nigerian credit infrastructure reform: A snapshot of progress and recommendations for next steps. <https://www.ifc.org/en/insights-reports/2024/nigerian-credit-infrastructure-reform-snapshot-of-progress-and-next-steps>
- Kane, G. C., Palmer, D., Phillips, A. N., Kiron, D., & Buckley, N. (2015). Strategy, not technology, drives digital transformation. MIT Sloan Management Review and Deloitte University Press. <https://sloanreview.mit.edu/projects/strategy-drives-digital-transformation/>
- National Bureau of Statistics, & Small and Medium Enterprises Development Agency of Nigeria. (2019). National survey of micro, small & medium enterprises (MSMEs) 2017. <https://nigerianstat.gov.ng/elibrary/read/966>
- Teece, D. J. (2007). Explicating dynamic capabilities: The nature and microfoundations of (sustainable) enterprise performance. Strategic Management Journal, 28(13), 1319-1350. <https://doi.org/10.1002/smj.640>

Recommended citation: CHIA TECH SOLUTIONS AND RESOURCES LIMITED. (2026). Empowering Nigerian MSMEs through Finance, Technology and Capability: The SCALE Framework for Sustainable Industrial Development. CHIATECH JOURNAL, 1(1), e018. DOI registration pending.

Identifier control: retain the eLocator; add only the registrar-confirmed DOI prefix and full DOI after successful deposit.